

## Research Interests

Natural Language Processing; Multi-party/Multi-modal Dialogue; Machine Learning; Deep Learning; Structured Prediction; Large Language Models; High-stakes Applications, such as AI for Education/Healthcare, etc.

## Education

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| 2015 – 2022 | <b>Doctor of Philosophy</b><br>Computer Science<br><i>University of Utah</i><br>Advisor: Vivek Srikumar<br>Dissertation: Inductive Biases for Deep Linguistic Structured Prediction<br>with Independent Factorization [19] |
| 2009 – 2012 | <b>Master of Science</b><br>Computer Science<br><i>Huazhong University of Science and Technology (HUST)</i>                                                                                                                |
| 2005 – 2009 | <b>Bachelor of Engineering</b><br>Information Security<br><i>Huazhong University of Science and Technology (HUST)</i>                                                                                                      |

## Research Experience

University of Oklahoma, Norman 2024 - Now

- I am an Assistant Professor in the School of Computer Science, leading the Natural Language Processing research group.

NSF-iSAT, University of Colorado Boulder, Boulder 2022 - 2024

- I was a **post-doctoral** researcher at NSF AI Institute for Student-AI Teaming (iSAT) with **Prof. James Martin and Prof. Martha Palmer**, Boulder NLP Groups, and Institute of Cognitive Science. My research mainly covers conversational modelling [8, 4, 18], speech-aware models [7, 6, 17], education AI on automatic feedbacks to teachers and tutors [1, 3], small group collaborative learning [6, 5].

Utah NLP Group, University of Utah, Salt Lake City 2015 - 2022

- I worked with **Prof. Vivek Srikumar** in Utah NLP Group. I study methods for predicting structured representations of natural language text, e.g., semantic parsing [12], dialogue system [14, 11, 9], self-supervised learning on database workloads [10].

Intern at AWS AI, Amazon Lex, Remote Summer 2020

- Our paper on schema-guided dialog got accepted to NAACL 2021 [9].

Intern at AWS AI, Amazon Lex, Seattle Summer 2019

- Cross framework meaning representation parsing, ranked 1st on AMR on CoNLL 2019 shared task [12].

Intern at Tencent, Wechat AI, Palo Alto Summer 2018

- End-to-End Goal-oriented Dialogue System [14], ranked 2nd in DSTC7 track1
- Rhetorically Controlled Poetry Generation[13] got accepted to ACL 2019.

CGCL Lab, HUST, Wuhan 2008 - 2012

- When I was a master student at HUST, I worked with Prof. Xia Xie and Prof. Hai Jin. My research are around large scale statistical computing, programming language [15, 16], system virtualization [21, 20]

## Teaching & Mentoring

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| Fall 2023      | Instructor | <b>Natural Language Processing (CSCI-LING 5832)</b> , U of Colorado Boulder |
| 2019-2020      | Mentor     | Bachelor Thesis (Tarun Sunkaraneni), U of Utah                              |
| Spring 2018    | TA         | Structured Prediction (CS 6355), U of Utah                                  |
| Fall 2016,2018 | TA         | Machine Learning (CS 6350), U of Utah                                       |
| Fall 2009      | Lecturer   | Algorithm and Data Structure, HUST                                          |
| 2006-2008      | Leader     | Algorithm&Game Team, Unique Studio, HUST                                    |

## Industrial Experience

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| Assistant Researcher at Sohu RDC Lab, Beijing                                                      | 2014 - 2015 |
| – Hadoop, Spark, Data migration, Data security, Distributed machine learning.                      |             |
| Senior Software Engineer at Zun Club, Beijing                                                      | 2013 - 2014 |
| – Heterogeneous data integration, Hotel recommendation system.                                     |             |
| Software Engineer at Baidu, Beijing                                                                | 2012 - 2014 |
| – Baidu Voice Assistant, Query analysis, Dialogue                                                  |             |
| – Mobile Search Anti-Attack Ecosystem, Speed optimization                                          |             |
| Intern at Alibaba, Hangzhou                                                                        | 2010 - 2011 |
| – KV Storage, MySQL, Database Replication, Real-time Computing, Distributed Pub/Sub Data Pipeline. |             |

## Honors & Awards

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| 2023 | iSAT Trainee Grant                                                               |
| 2019 | CoNLL Shared Task, MRP, Top 1 on AMR                                             |
| 2019 | DSTC7, Top 2 on track 1                                                          |
| 2010 | VMware Cloud Computing Innovation Cup, Top 50                                    |
| 2009 | Google Android Innovative Idea Sharing Award                                     |
| 2007 | “Computer World” Magazine Scholarship(50 students per year in China)             |
| 2007 | Microsoft Imagine Cup in Visual Gaming Contest(Top 2 in China/Top 18 World-wide) |
| 2006 | HUST ACM Programming Contest, Top 3                                              |

## Publications

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- [1] **Jie Cao**, Abhijit Suresh, Jennifer Jacobs, Charis Clevenger, Amanda Howard, Chelsea Brown, Brent Milne, Tom Fischhaber, Tamara Sumner, and James H. Martin. “Enhancing Talk Moves Analysis in Mathematics Tutoring through Classroom Teaching Discourse”. *The 31st International Conference on Computational Linguistics (COLING 2025)*. 2025.
  - [2] Zhimin Li, Shusen Liu, Xin Yu, Kailkhura Bhavya, **Jie Cao**, Diffenderfer James Daniel, Peer-Timo Bremer, and Valerio Pascucci. “Understanding Robustness Lottery”: A Geometric Visual Comparative Analysis of Neural Network Pruning Approaches”. *IEEE Transactions on Visualization and Computer Graphics* (2024).
  - [3] Baptiste Moreau-Pernet, Tian Yu, Sandra Sawaya, Peter Foltz, **Jie Cao**, Brent Milne, and S. Thomas Christie. “Classifying Tutor Discursive Moves at Scale in Mathematics Classrooms with Large Language Models”. *Learning @ Scale 2024*. 2024.
  - [4] Jon Z Cai, Brendan King, Margaret Perkoff, Shiran Dudy, **Jie Cao**, Marie Grace, Natalia Wojarnik, Ananya Ganesh, James H Martin, Martha Palmer, et al. “Dependency Dialogue Acts–Annotation Scheme and Case Study”. *The 13th International Workshop on Spoken Dialogue Systems Technology* (2023).

- [5] **Jie Cao**, Rachel Dickler, Marie Grace, Alessandro Roncone, Leanne Hirshfield, Marilyn Walker, and Martha Palmer. “Designing an AI Partner for Jigsaw Classrooms”. *Workshop on Language-Based AI Character Interaction with Children* (2023).
- [6] **Jie Cao**, Ananya Ganesh, Jon Cai, Rosy Southwell, E Margaret Perkoff, Michael Regan, Katharina Kann, James H Martin, Martha Palmer, and Sidney D’Mello. “A Comparative Analysis of Automatic Speech Recognition Errors in Small Group Classroom Discourse”. *Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization*. 2023, pp. 250–262.
- [7] Ananya Ganesh, **Jie Cao**, E. Magerate Perkoff, Rosy Southwell, Martha Palmer, and Katharina Kann. “Mind the Gap between the Application Track and the Real World”. *Proceedings of the 61th Annual Meeting of the Association for Computational Linguistics, 2023* (2023).
- [8] E. Margaret Perkoff, Abhidip Bhattacharyya, Jon Cai, and **Jie Cao**. “Comparing Neural Question Generation Architectures for Reading Comprehension”. *18th Workshop on Innovative Use of NLP for Building Educational Applications, 2023* (2023).
- [9] **Jie Cao** and Yi Zhang. “A Comparative Study on Schema-Guided Dialogue State Tracking”. *Proceedings of NAACL 2021*. 2021.
- [10] Debjyoti Paul\*, **Jie Cao\***, Feifei Li, and Vivek Srikumar. “Database Workload Characterization with Query Plan Encoders”. *Proc. VLDB Endow.* 15.4 (2021), pp. 923–935. issn: 2150-8097. doi: [10.14778/3503585.3503600](https://doi.org/10.14778/3503585.3503600). url: <https://doi.org/10.14778/3503585.3503600>.
- [11] **Jie Cao**, Michael Tanana, Zac Imel, Eric Poitras, David Atkins, and Vivek Srikumar. “Observing Dialogue in Therapy: Categorizing and Forecasting Behavioral Codes”. *Proceedings of ACL 2019*. Florence, Italy, 2019.
- [12] **Jie Cao**, Yi Zhang, Adel Youssef, and Vivek Srikumar. “Amazon at MRP 2019: Parsing Meaning Representations with Lexical and Phrasal Anchoring”. *Proceedings of the Shared Task on MRP at the CoNLL 2019*. 2019.
- [13] Zhiqiang Liu, Zuohui Fu, **Jie Cao**, Gerard de Melo, Yik-Cheung Tam, Cheng Niu, and Jie Zhou. “Rhetorically Controlled Encoder-Decoder for Modern Chinese Poetry Generation”. *Proceedings of ACL 2019*. Florence, Italy, 2019.
- [14] Shuo Sun\*, Yik-Cheung Tam\*, **Jie Cao\***, Can-Xiang Yan, Zuohui Fu, Cheng Niu, and Jie Zhou. “End-to-end Gated Self-attentive Memory Network for Dialog Response Selection”. *Proceedings of AAAI, DSTC7 Workshop*. 2019.
- [15] Xijiang Ke, Hai Jin, Xia Xie, and **Jie Cao**. “A distributed SVM method based on the iterative MapReduce”. *International Conference on Semantic Computing(ICSC)*. IEEE. 2015.
- [16] Xia Xie, **Jie Cao**, Hai Jin, Xijiang Ke, and Wenzhi Cao. “JRBridge: A framework of large-scale statistical computing for R”. *Asia-Pacific on Services Computing Conference (APSCC)*. IEEE. 2012.

## Preprints or Under-Preparation

- [17] **Jie Cao**, Jon Cai, Viet Anh, Rosy Southwell, Jacob Whitehill, Jeff Flanigan, James Martin, and Martha Palmer. “Speech-aware AMR Parsing”. 2023.
- [18] **Jie Cao**, Jon Cai, E Margaret Perkoff, James Martin, and Martha Palmer. “Dialogue Synthesis on Reinforcing Small Group Discussion with Peer Role Conversational Agents”. 2023.

## Thesis & Patents

- [19] **Jie Cao**. “Inductive Biases for Deep Linguistic Structured Prediction with Independent Factorization”. University of Utah, 2022.
- [20] **Jie Cao**, Xia Xie, and Jin Hai. “A real-time scheduling system on embeded virtualization”. Patent ZL201110410689.1 (CN). 2011.
- [21] **Jie Cao**. “Cache Performance Evaluation on Xen Virtualization Platform”. Huazhong University of Science and Techonology, 2009.

## Talks

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- 09/2024, Invited Talks on "History of NLP" at OU AI/ML Club.
- 03/2024, Invited Talks at Emory, Georgia State University, Northern Illinois University, University of Colorado Boulder(CompSem).
- 12/2022, Invited Talk on Database Workload Characterization work at Microsoft's Gray Systems Lab.
- 05/2021, Guest Lecturer @ University of Idaho: CS 501 Seminar: Contemporary Issues, 'Inductive Biases in Deep Linguistic Structured Prediction'
- 12/2020, Talk on EMNLP'2020 Watch Party@Amazon Lex, 'Task-oriented Conversational Semantic Parsing'
- 11/2019, MRP 2019, 'Amazon at MRP 2019: Parsing Meaning Representations with Lexical and Phrasal Anchoring'

## Software

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Other code for papers can be found in <https://github.com/utahnlp> and <https://github.com/mlciv>

- Therapist-Observer, a family of neural components for various hierarchical dialogue models for motivational interviewer, described in our ACL'19 [11].
- LAPA-MRP, our open-sourced system submission to MRP 2019 shared task at CoNLL 2019. Our model ranked 1st in the AMR subtask, 5th in UCCA, 6th in PSD and 7th in DM.
- sgd, which implements a family of neural components for schema-guided dialog used in our NAACL'2021 paper [9], including encoder architectures, description styles, supplementary training and so on.
- dt-proxy, A nodejs-based token proxy to support distributed data copy between secure and insecure hadoop clusters.
- renjin, a JVM-based interpreter for the R language. <https://github.com/bedatadriven/renjin>

## Academic Service

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### Program Committee Member/Reviewer

- International Conference on Artificial Intelligence in Education (AIED), 2023,2024
- International Conference on Educational Data Mining (EDM), 2024
- Conference on Language Modeling(COLM), 2024
- Conference of the Association for the Advancement of Artificial Intelligence (AAAI), 2020–24
- Annual meeting of the Association of Computational Linguistics (ACL), 2020-24
- International Conference on Computational Linguistics (COLING), 2020,22,24
- Conference on Computational Natural Language Learning (CoNLL), 2021-2023
- Conference of the European Chapter of the ACL (EACL), 2021
- Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020-2023
- North American Chapter of the ACL (NAACL), 2021
- Shared Task on Cross-Framework Meaning Representation Parsing at the 2019 Conference on Natural Language Learning (MRP'2019@CoNLL)
- ACL Rolling Review, 2021-24
- Workshop on Innovative Use of NLP for Building Educational Applications (BEA), 2023

- Workshop on NLP for Conversational AI (NLP4ConvAI), 2023
- Workshop on NLP for Indigenous Languages of the Americas (AmericasNLP), 2023
- Workshop on Speech and Language Technology in Education (SLaTE), 2023
- Workshop on Queer in AI, 2023